WHEN WILL A VLM FAIL ON MEDICAL VQA? IMAGE-FREE RISK PREDICTION AND SHARED ERROR STRUCTURE

Marouane Tliba[1], *Mohamed Amine Kerkouri*[2,1], *Youssra Taifour*[1], *Benyahyia Mohamed Oussama*[1], Emadeldeen Hamdan[3], Omid Halimi Milani[3], Zakaria Selam[1], Nour Abouread[4], Aladine Chetouani[1], Ahmet Enis Cetin[3], Azeddine Beghdadi[1]

[1]Université Sorbonne Paris Nord; [2]F-Initiatives; [3]University of Illinois Chicago; [4]University of Dubai

ABSTRACT

Vision–language models (VLMs) can produce plausible but incorrect answers on medical visual question answering even under ordinary, intact inputs. We study whether the failure of an already generated answer can be predicted before any test-time intervention. Across five frozen VLMs on VQA-RAD and SLAKE, a predictor using only the question and generated answer reaches AUROC 0.77–0.94, while frozen image features provide little additional failure-predictive information and standard confidence signals are generally weaker. Failures also exhibit strong cross-model structure: approximately one quarter of test items are answered incorrectly by all five VLMs, far above the rate expected under independent errors. Counterfactual image replacement and degradation confirm that the models do use visual information, showing that failure predictability is not simply image blindness. Although per-item oracle correction reveals substantial headroom, inexpensive test-time interventions do not recover it reliably; in contrast, the same risk score supports effective selective answering. Our results reveal shared item- and question–answer structure in VLM failures on medical VQA and distinguish visual dependence, failure detectability, and correctability. Code source are available at: `https://github.com/mtliba/VLM-ImFreeRisk.git`.

***Index Terms*—** medical vision–language models, failure prediction, uncertainty estimation, selective prediction, medical visual question answering

1. INTRODUCTION

Medical vision–language models (VLMs) increasingly answer questions about radiographs and other medical images, yet their short responses can remain plausible and wrong. Benchmarks such as VQA-RAD [1] and SLAKE [2] measure average task performance, but do not tell us whether a particular generated answer should be trusted. This motivates a complementary reliability question: *after a VLM has processed an ordinary, intact image–question pair and generated an answer, can we predict whether that answer is wrong?*

Figure 1 illustrates our setting. The VLM receives the original image and question and produces its normal answer.

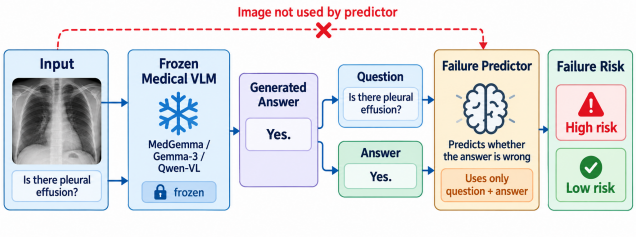


Fig. 1. **Intact-input failure prediction.** The frozen VLM receives the original image and question. A downstream predictor estimates whether its generated answer is wrong using only the question and answer; the image is never provided to the failure predictor.

A separate downstream predictor then estimates failure risk using only the question and generated answer; it never observes the image. This distinction is important. Prior VQA work has documented language priors, dataset shortcuts, and weak visual dependence [3, 4, 5, 6, 7, 8], while recent medical-VLM evaluations remove, replace, or invalidate visual evidence to study grounding and behavior under broken evidence [9, 10]. Our target is different: *per-answer failure prediction under intact inputs*. We ask whether the answer produced during normal operation is likely to be wrong before deciding whether to accept, defer, or modify it.

Across five frozen VLMs on VQA-RAD and SLAKE, we find that this failure signal is surprisingly strong: question–answer features yield AUROC 0.77–0.94, whereas frozen image features provide little failure-predictive information and standard confidence signals are generally weaker. The effect is not explained solely by repeated templates: grouping identical SLAKE questions across folds reduces, but does not eliminate, failure predictability. Instead, failures exhibit substantial shared structure: roughly one quarter of test items are failed by all five VLMs, far above the rate expected under independent errors. Importantly, counterfactual image replacement and degradation confirm that the models still use visual information. Predictable failure is therefore not simply image blindness; difficult items can be image-sensitive while remaining systematically error-prone across models.

This distinction also determines what the risk signal is

useful for. Inexpensive test-time interventions; including re-prompting, image enhancement, zooming, learned routing, consensus, and self-verification—leave substantial oracle correction headroom unrecovered and can introduce harmful answer changes [11, 12, 13]. In contrast, the same risk score supports effective selective answering [14]. The resulting picture separates three notions that are often conflated: *visual dependence*, *failure detectability*, and *correctability*.

Our contributions are threefold: We (i) formulate image-free per-answer failure prediction under intact medical-VQA inputs and demonstrate strong question–answer-based prediction across five VLMs; (ii) identify shared item- and benchmark-level failure structure while showing that it is not simply image blindness; and (iii) show that failure detectability does not imply correctability, with selective answering providing a more reliable use of the risk signal than inexpensive test-time correction.

2. EXPERIMENTAL SETUP

Benchmarks and models. We evaluate on VQA-RAD [1] and SLAKE-English [2]. VQA-RAD contains 451 test QA pairs over 203 images; 15% of the training images are reserved for validation. SLAKE provides 4,919/1,053/1,061 train/validation/test pairs. We study five frozen VLMs: MedGemma-4B-IT [15], Gemma-3-4B-IT [16], Qwen2.5-VL-7B/3B-Instruct [17], and Qwen2-VL-2B-Instruct [18], all with greedy decoding and a common short-answer prompt. For generalization, we also evaluate failure prediction on PathVQA [19], PMC-VQA [20], and OmniMedVQA [21] using the same protocol.

Failure prediction. For image I, question q, reference $y^\star$, and answer $\hat{y} = f(I, q)$, we define

$$e = 1[S(\hat{y}, y^\star) < 1],$$

where S is Metric A: normalized exact matching for yes/no answers and token-F1 for open answers; $S = 1$ denotes a fully correct answer. The VLM always receives the intact image; only the downstream predictor may be image-free. Our main predictor (**Q+A**) is logistic regression on frozen BiomedCLIP [22] question and answer embeddings. We compare it with image and image+Q+A features, token log-probability, self-consistency [23, 24], agreement under re-prompting, enhancement, and zoom, and a linear hidden-state probe [25, 26].

Evaluation. Main results train learned predictors on train plus validation and evaluate on the official test split. We report AUROC for failure prediction and exact accuracy for selective answering, with 1,000 bootstrap replicates. We additionally use image-grouped five-fold cross-validation and a SLAKE template-held-out variant that keeps identical normalized questions in the same fold.

3. PREDICTING FAILURES WITHOUT THE IMAGE

Question–answer structure strongly predicts failure. Table 1 summarizes failure prediction across all five VLMs.

Table 1. **Failure prediction under intact inputs.** Acc. and Sel.@50 are exact-answer accuracy (%); other columns report AUROC for incorrect-answer prediction.

	VLM	Acc.	Img.	Log-p	Samp.	Pert.	Q+A	Probe	Sel.@50
VQA-RAD	MedGemma-4B	59.2	.49	.67	.62	.71	**.79**	.85	81
	Gemma-3-4B	35.9	.52	.52	.56	.64	**.77**	.85	57
	Qwen2.5-VL-7B	55.2	.49	.80	.76	.73	**.78**	.82	76
	Qwen2.5-VL-3B	49.2	.56	.65	.74	.70	**.78**	.81	69
	Qwen2-VL-2B	42.4	.54	.71	.72	.64	**.78**	.83	63
SLAKE	MedGemma-4B	60.2	.58	.59	.58	.67	**.91**	.96	91
	Gemma-3-4B	32.0	.58	.49	.56	.66	**.94**	.94	63
	Qwen2.5-VL-7B	52.8	.57	.68	.77	.73	**.90**	.95	84
	Qwen2.5-VL-3B	55.1	.56	.78	.78	.70	**.88**	.93	84
	Qwen2-VL-2B	43.5	.53	.75	.73	.67	**.91**	.93	77

A lightweight linear predictor using only the question and the VLM's generated answer reaches AUROC 0.77–0.79 on VQA-RAD and 0.88–0.94 on SLAKE across model families and scales. These values are substantially above chance despite the predictor never observing the image. Thus, after a VLM has processed an ordinary intact input, the identity of many of its failures is already predictable from the resulting question–answer pair.

Image features add little information about which answers fail. Frozen image embeddings alone are weak failure predictors (0.49–0.58 AUROC). In the four primary settings, augmenting Q+A with the corresponding image embedding changes AUROC only marginally: 0.789→0.779 and 0.776→0.754 on VQA-RAD, and 0.906→0.912 and 0.901→0.915 on SLAKE. This result concerns *failure prediction*: it does not imply that the VLM itself ignores the image when producing its answer. We examine that distinction explicitly in Sec. 4.

The Q+A signal is not simply conventional model confidence. As shown in Table 1, Q+A is generally competitive with or stronger than token log-probability, sampling self-consistency, and perturbation consistency, while requiring no additional VLM generations. The difference is particularly pronounced on SLAKE, where Q+A reaches 0.88–0.94 AUROC whereas the black-box confidence signals are markedly less consistent across models. A linear probe on the VLM's final hidden representation is stronger still, reaching 0.81–0.85 AUROC on VQA-RAD and 0.93–0.96 on SLAKE. Failure risk is therefore represented internally, but a large fraction of the useful signal is already observable from the generated question–answer pair.

Exact template reuse explains only part of the effect. SLAKE contains substantial question repetition, raising the possibility that the predictor simply memorizes recurring templates. We therefore repeat cross-validation while forcing identical normalized question strings into the same fold. Q+A AUROC decreases from 0.93/0.92 to 0.84/0.79 for MedGemma/Qwen, respectively, but remains well above chance; the corresponding VQA-RAD results change little.

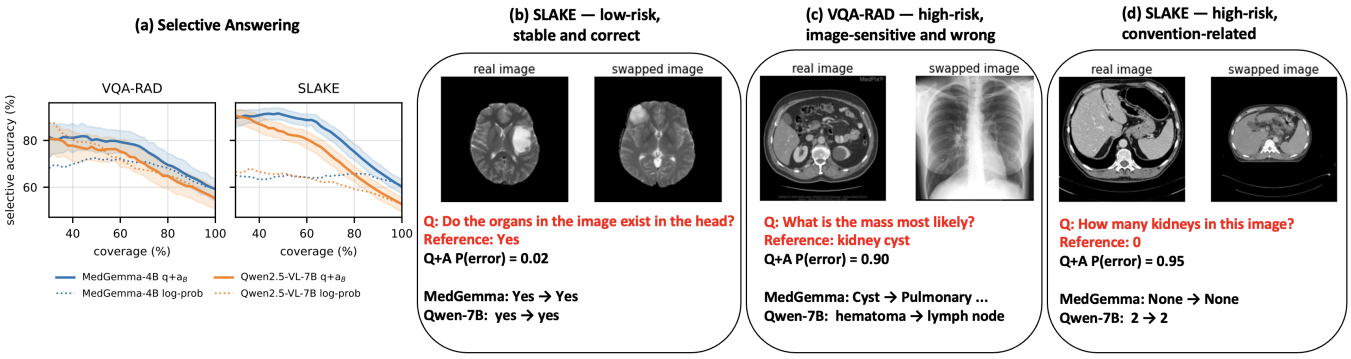


Fig. 2. **Selective answering and representative failure modes.** **(a)** Risk–coverage curves using Q+A risk (solid) and token log-probability (dotted). **(b)** Low-risk, image-insensitive correct case. **(c)** High-risk, image-sensitive incorrect case. **(d)** High-risk convention-related failure. Predictable failure is therefore not equivalent to image blindness.

Template recurrence therefore amplifies predictability on SLAKE, but cannot by itself account for the observed signal. This motivates examining whether failures instead exhibit broader item- and benchmark-level structure.

4. SHARED STRUCTURE OF PREDICTABLE FAILURES

Why is failure predictable without re-examining the image? We examine three complementary sources of structure: visual dependence, cross-model overlap, and benchmark regularities.

Predictable failure is not simply image blindness. Table 2 confirms that the evaluated VLMs make substantial use of visual information. Removing the image sharply reduces Metric-A score, and progressive blur or downsampling similarly drives performance toward the no-image floor. Yet visual sensitivity is not aligned with correctness. Under image replacement, 52–69% of answers remain unchanged for the two primary VLMs, and these image-insensitive answers are consistently more accurate than answers that change. Moreover, high-risk cases are generally *more* image-sensitive, not less. Thus, the image-free risk signal cannot be explained by a simple failure mode in which incorrect answers arise because the model ignores the image.

The same difficult items fail across VLMs. Pairwise error correlations across the five models range from $\phi = 0.32$ to 0.60. More strikingly, 26.8% of VQA-RAD and 25.7% of SLAKE items are failed by all five VLMs, compared with only 3.4% and 3.2% expected under independent errors (Table 2). This shared hard core persists across specialization: MedGemma and its Gemma-3 base are jointly wrong on 35.5% of VQA-RAD and 38.0% of SLAKE items, although medical tuning repairs 28.6% and 30.0% of the base model's cases. A failure predictor trained on one VLM also transfers to another at AUROC 0.71–0.83. Together, these results indicate a substantial item-level component of failure rather than purely model-specific uncertainty.

Benchmark structure explains part, but not all, of the shared difficulty. On SLAKE, 43–47% of errors from the two main VLMs occur on recurring (question, answer) pairs that are wrong in at least 90% of their occurrences, often involving answer conventions, count formats, compound labels, or knowledge-oriented questions. Exact repetition is not the whole explanation: the template-held-out analysis in Sec. 3 remains strongly predictive after identical questions are grouped across folds.

VQA-RAD provides a complementary case. Exact question–answer repetition accounts for only a small fraction of its errors, yet failure remains predictable at about 0.78 AUROC across all five models. Broader question form is informative, with localization questions consistently harder than presence or imaging-plane questions. The relevant structure is therefore broader than template memorization and reflects systematic item- and question–answer difficulty.

Benchmark construction moderates the effect. The pattern is not universal. Failure prediction remains strong on PathVQA [19] and OmniMedVQA [21], whereas PMC-VQA [20] provides an informative counterexample: Q+A AUROC falls to 0.59–0.65 and the relation between image sensitivity and correctness reverses. This argues against a universal image-blindness explanation and instead points to benchmark construction as an important moderator.

5. CORRECTION OR ABSTENTION?

Failure detection and failure correction are distinct problems. Estimating that the current answer is likely wrong does not identify which alternative answer, prompt, or visual transformation will improve it.

Correction headroom exists, but is difficult to realize. We evaluate structured re-prompting, contrast enhancement, and multi-crop zoom together with KEEP. The per-item oracle exceeds the best fixed intervention by 4.5–10.4 Metric-A points in the four primary settings, confirming that useful corrections exist. However, KEEP is already oracle-optimal for 84–92%

Table 2. **Structure of predictable failures.** Real/No-img. are Metric-A scores; Swap unch. is answer stability under image replacement and Unch./Ch. the corresponding conditional scores. All-5/Ind. compares observed five-model joint failure with independence; $\bar{\phi}$ is mean pairwise error correlation.

	VLM	Real	No-img.	Swap unch.	Unch./Ch.	All-5/Ind.	$\bar{\phi}$
VQA-RAD	MedGemma-4B	64.3	33.3	51.7	71.9/56.1	26.8/3.4	.45
	Qwen2.5-VL-7B	58.8	33.4	58.3	67.8/46.2		
SLAKE	MedGemma-4B	65.1	24.8	59.3	73.1/53.4	25.7/3.2	.50
	Qwen2.5-VL-7B	56.4	23.1	69.3	62.2/43.4		

of items, and every intervention both repairs and damages answers. The main difficulty is therefore not generating alternatives, but deciding when an alternative is genuinely better.

We also test learned gating and action prediction, intervention consensus, and anchored self-verification. As summarized in Table 3, these methods recover only part of the oracle headroom. The strongest result is on SLAKE/Qwen, where gating raises Metric-A score from 56.4 to 62.8, but approximately 9% of selected changes are still harmful. On VQA-RAD, gating is essentially neutral or detrimental despite strong failure prediction. Consensus provides little independent corrective evidence, while anchored verification changes very few answers. In short, *knowing that an answer is risky is easier than knowing how to replace it*.

Selective answering provides a more reliable use of failure risk. Rather than modifying a high-risk answer, we rank test examples by Q+A failure probability and abstain on the highest-risk fraction. At 70% coverage, exact accuracy of the retained answers rises to 76/70/84/74% for VQA-RAD/MedGemma, VQA-RAD/Qwen, SLAKE/MedGemma, and SLAKE/Qwen, respectively. At 50% coverage, it reaches 81/76/91/84%, compared with full-coverage accuracies of 59/55/60/53%. Thus, the same signal that is insufficient for reliable answer replacement remains highly useful for deciding which answers to retain. Figure 2(a) shows the corresponding risk–coverage behavior across the full operating range: accuracy rises as high-risk answers are removed, and Q+A provides a stronger ranking than token log-probability in three of the four primary settings. The qualitative examples in Fig. 2(b–d) illustrate complementary cases: a low-risk answer can remain correct despite image replacement, whereas high-risk answers can be visually sensitive yet wrong or reflect recurring answer conventions. These examples reinforce the distinction established in Sec. 4: failure predictability is not equivalent to image blindness, and failure detectability does not imply correctability.

Overall, the practical implication is asymmetric: the risk score is useful for deciding *when to answer*, but does not reliably determine *how to correct* a flagged response.

Table 3. **Correction versus selective answering.** Base, Fixed, Gate, and Oracle report Metric-A score; Fixed is the best validation-selected intervention, Gate the learned gate+action policy, and Oracle the per-item best action. Sel.@50 is exact accuracy (%) on the 50% lowest-risk answers ranked by Q+A.

Setting	Base	Fixed	Gate	Oracle	Sel.@50
VR / MG	64.3	62.9	63.1	70.4	81
VR / Q7	58.8	60.5	58.8	68.2	76
SL / MG	65.1	66.3	67.4	70.8	91
SL / Q7	56.4	59.2	62.8	69.6	84

6. DISCUSSION AND CONCLUSION

This work studies a reliability question distinct from visual grounding: whether the failure of a VLM answer on medical VQA can be predicted after the model has processed an ordinary, intact input. Across VQA-RAD and SLAKE, question–answer features provide a strong failure signal, while frozen image features add little information about *which* answers will fail. This predictability is not explained solely by template reuse: it remains substantial when identical SLAKE questions are grouped across folds, and VQA-RAD exhibits similar structure despite little exact repetition.

The resulting picture is more nuanced than simple image blindness. Removing or degrading the image substantially reduces performance, so the VLMs clearly use visual information. Yet failure risk is also strongly structured at the item and benchmark level: image-insensitive answers are often more accurate, high-risk cases can remain image-sensitive, and roughly one quarter of test items are failed by all five VLMs, far above the rate expected under independent errors. Medical specialization improves overall accuracy but leaves a substantial shared hard core. More broadly, the variation observed across benchmarks suggests that predictable failure depends in part on benchmark construction rather than being a universal property of VLMs.

These findings separate three notions that are often conflated: *visual dependence*, *failure detectability*, and *correctability*. A VLM may use the image while its failures remain predictable from the question–answer pair; conversely, detecting that an answer is risky does not reveal which alternative will be better. This is reflected by the gap between the per-item correction oracle and practical routing, consensus, and verification, whereas the same risk score supports effective selective answering. In this setting, knowing *when not to trust* an answer is easier than knowing *how to repair* it.

Our conclusions are limited to short-answer medical VQA and should not be interpreted as evidence that VLMs generally ignore images. Rather, they show that reliability evaluation should distinguish visual dependence, failure predictability, and correctability. For radiology VQA, shared item- and question–answer structure provides a useful reliability signal and a practical diagnostic before attempting test-time correction.

7. REFERENCES

[1] Jason J. Lau et al., "A dataset of clinically generated visual questions and answers about radiology images," *Scientific Data*, vol. 5, pp. 180251, 2018.

[2] Bo Liu et al., "Slake: A semantically-labeled knowledge-enhanced dataset for medical visual question answering," 2021.

[3] Yash Goyal et al., "Making the V in VQA matter: Elevating the role of image understanding in visual question answering," in *IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017.

[4] Aishwarya Agrawal, Dhruv Batra, Devi Parikh, and Aniruddha Kembhavi, " Don't Just Assume; Look and Answer: Overcoming Priors for Visual Question Answering ," in *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Los Alamitos, CA, USA, June 2018, pp. 4971–4980, IEEE Computer Society.

[5] Rémi Cadène et al., "RUBi: Reducing unimodal biases for visual question answering," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[6] Robert Geirhos et al., "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[7] Lin Chen et al., "Are we on the right way for evaluating large vision-language models?," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024, arXiv:2403.20330.

[8] Shengbang Tong et al., "Eyes wide shut? Exploring the visual shortcomings of multimodal LLMs," in *IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2024.

[9] Anas Zafar et al., "Do medical vision language models actually see? a counterfactual grounding framework and hard-negative contrastive training for visually-reliant medical vlms," *arXiv preprint arXiv:2607.03647*, 2026.

[10] Hanqi Jiang et al., "Medvigil: Evaluating trustworthy medical vlms under broken visual evidence," 2026.

[11] Aman Madaan et al., ," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[12] Jie Huang et al., "Large language models cannot self-correct reasoning yet," in *Int. Conf. Learning Representations (ICLR)*, 2024, arXiv:2310.01798.

[13] Ryo Kamoi et al., "When can LLMs actually correct their own mistakes? A critical survey of self-correction of LLMs," *Transactions of the Association for Computational Linguistics*, vol. 12, pp. 1417–1440, 2024.

[14] Yonatan Geifman and Ran El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[15] Andrew Sellergren et al., "Medgemma technical report," 2026.

[16] Gemma Team, Aishwarya Kamath, et al., "Gemma 3 technical report," 2025.

[17] Shuai Bai et al., "Qwen2.5-VL technical report," *arXiv preprint arXiv:2502.13923*, 2025.

[18] Peng Wang et al., "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," *arXiv preprint arXiv:2409.12191*, 2024.

[19] Xuehai He et al., "Pathvqa: 30000+ questions for medical visual question answering," 2020.

[20] Xiaoman Zhang et al., "PMC-VQA: Visual instruction tuning for medical visual question answering," *arXiv preprint arXiv:2305.10415*, 2023.

[21] Yutao Hu et al., " OmniMedVQA: A New Large-Scale Comprehensive Evaluation Benchmark for Medical LVLM ," in *2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2024, pp. 22170–22183.

[22] Sheng Zhang et al., "BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs," *arXiv preprint arXiv:2303.00915*, 2023.

[23] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou, "Self-consistency improves chain of thought reasoning in language models," in *Int. Conf. Learning Representations (ICLR)*, 2023, arXiv:2203.11171.

[24] Potsawee Manakul, Adian Liusie, and Mark J. F. Gales, "SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models," in *Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[25] Saurav Kadavath et al., "Language models (mostly) know what they know," *arXiv preprint arXiv:2207.05221*, 2022.

[26] Amos Azaria and Tom Mitchell, "The internal state of an LLM knows when it's lying," in *Findings of EMNLP*, 2023, arXiv:2304.13734.